

Product Specification

SPECIFICATION
FOR
APPROVAL

() Preliminary Specification
(●) Final Specification

Title	
-------	--

Customer	Lenovo
Part Number	SD68E26070

SUPPLIER	CSOT
MODEL	PP8807DB2-4
Version	V04

SIGNATURE	DATE
/	
/	
/	

Please return 1 copy for your confirmation with your signature and comments.

APPROVED BY	DATE
REVIEWED BY	

Product Specification

CONTENT LIST

No	ITEM	Page
1	COVER	1
2	CONTENTS	2
3	RECORD OF REVISIONS	3
4	GENERAL DESCRIPTION	4
5	ABSOLUTE MAXIMUM RATINGS	12
7	ELECTRICAL SPECIFICATIONS	13
8	OPTICAL CHARACTERISTICS	29
9	MECHANICAL CHARACTERISTICS	36
10	RELIABILITY TEST	41
11	PACKING	42
12	Incoming Inspection Standard	43
13	Checklist	46

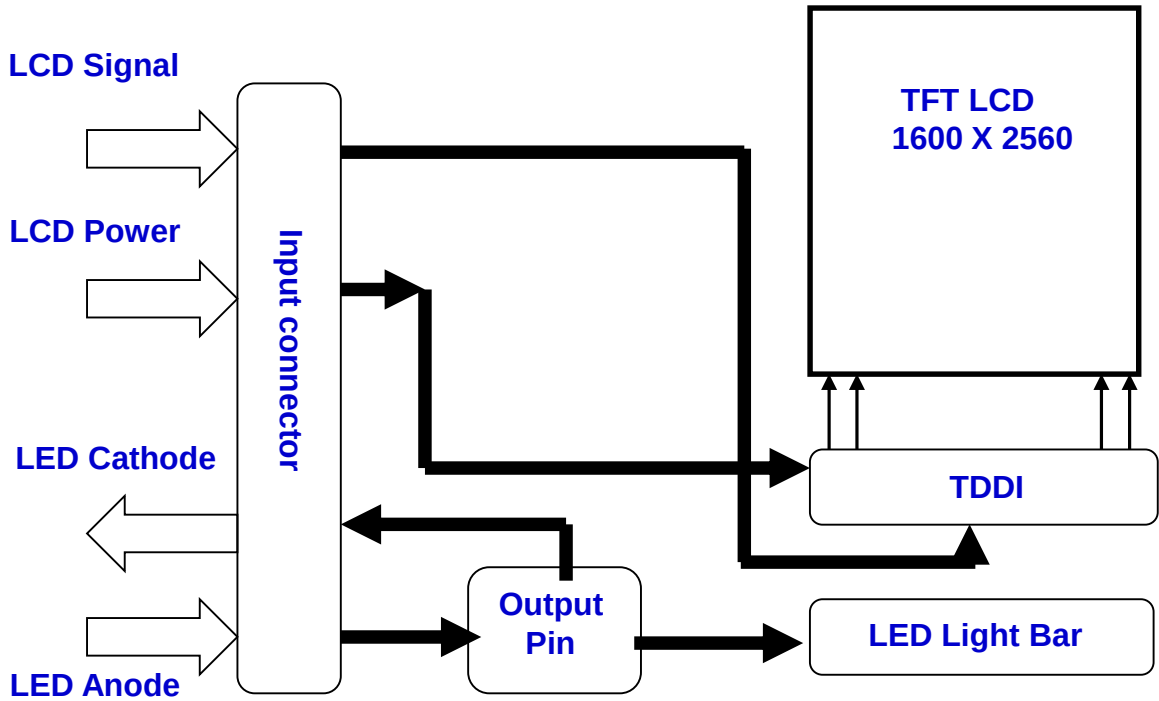
RECORD OF REVISIONS

3 / 41

Product Specification

1-1. GENERAL DESCRIPTION

Block Diagram



Features

PP8807DB2-4 is 8.8" color TFT-LCD (Thin Film Transistor Liquid Crystal Display) with low blue light and full in-cell module composed of LCD panel, MIPI C-phy driver ICs, control circuit ,backlight and cover glass. By applying 8 bit digital data, 1600×RGB (3) ×2560, 16.7M-color images are displayed on the 8.8" screen.

Product Specification

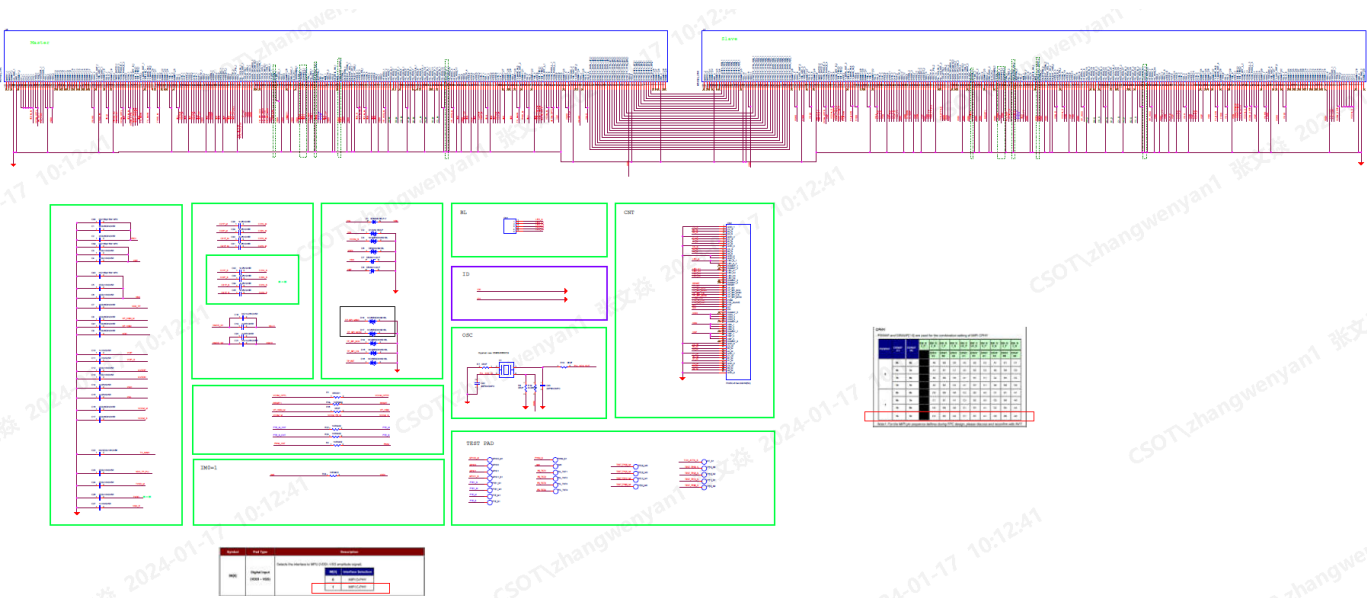
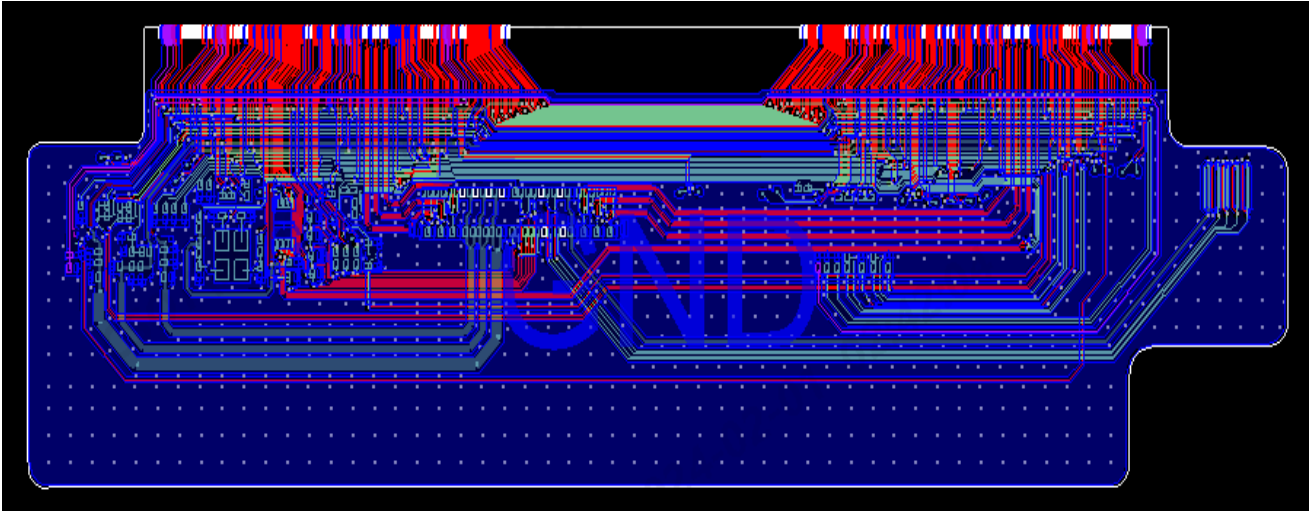
1-2. General Spec

No	Item	Specification	unit	Remark
1	Screen Size	8.8	inch	
2	Active Area	118.464x189.5424	mm	
3	Panel Size	121.464*195.2424	mm	
4	Outline Dimension	125.96*205.04*2.15	mm	
5	Display Resolution	1600×RGB×2560	pixel	
6	Pixel Pitch	74.04*74.04	um	
7	Display Method	LTPS	-	a-si/LTPS
8	Display Mode	FFS	-	TN/VA/IPS
9	Display Color	16.7M	-	
10	Color Gamut	Typ 96%, Min 91%	%	DCI-P3
		Typ 96%, Min 91%	%	NTSC
11	Luminance	Typ 750, Min 640	nit	typ
12	Contrast Ratio	Typ 1500, Min 1000	-	typ
13	Viewing Angle	U/D/L/R : Min80/80/80/80	°	CR>10(U/D/L/R)
14	Pol Surface Treatment	HC	-	HC
15	Weight	98	g	max
16	D-IC	NT36523NEH-DP/3YA	-	
17	Inversion Method	Column		
18	LED Q'ty	36	ea	9*4
19	Logic Power Consumption	585mW @165Hz	mw	white pattern&with TP
		2306mW @165Hz	mW	Subpixel onoff 255 pattern&with TP
20	BL Power Consumption	2392.9mW max @Max brightness	mW	BL Current:23mA
21	AES pen	Lpp	-	-
22	DP	165/144/120/90/60/30	Hz	
23	TP 报点率	Hand:165/144/90 TP 报点率 : 165/144/90	Hz	
		Pen+Hand:120/60/30 TP 报点率 : 240		

Product Specification

1-3-1. FPCA Gerber/Layout and Schematic Diagram

方案：连接器左移 12.8+1mm



Product Specification

1-3-2. FPC Pin Assignment Master

Please pay attention that IC bump down(TFT glass up and C/F glass down)

270	DUMMY	180	VDD	90	VG_HSSI
269	AG_GND	179	VDD_TP	89	VP_HSSI
268	GND	178	VDD_TP_PLL	88	VP_HSSI
267	DUMMY	177	VSS	87	VG_HSSI
266	CT1	176	PLL_XCK_IN	86	VDD
265	CT2	175	PLL_XCK_OUT	85	VSS
264	CT3	174	TCH_ATTEN	84	VDDI
263	CT4	173	TCH_IO_SDA	83	VDD_TP
262	CT5	172	TCH_IO_SCL	82	AVEE
261	CT6	171	FLASH_SPI_MISO	81	AVSS
260	TPDO	170	FLASH_SPI_MOSI	80	VCIP
259	TPDE	169	FLASH_SPI_SCK	79	AVDD
258	FOGTEST4	168	FLASH_SPI_CSN	78	CVSS
257	FOGTEST3	167	TCH_IO_SCK	77	C22N
256	VSS	166	TCH_IO_MOSI	76	C22P
255	VSS	165	TCH_IO_CSN	75	VGH
254	VDDI	164	TCH_IO_MISO	74	VGH
253	VDDI	163	TP_OPT	73	CVSS
252	DUMMY	162	GPIO[0]	72	AVEE
251	DUMMY	161	TP_MS	71	AVDD
250	DUMMY	160	VDDI_DC	70	C21N
249	DUMMY	159	VSS	69	C21P
248	DUMMY	158	VDDI	68	DUMMY
247	DUMMY	157	DUMMY	67	AVDD
246	DUMMY	156	RESX	66	DUMMY
245	DUMMY	155	ENB_CASC	65	DUMMY
244	DUMMY	154	LCD_LR	64	VDD
243	DUMMY	153	VDDIDUM[0]	63	DUMMY
242	DUMMY	152	TP_EN	62	TX_MAIN_IN
241	DUMMY	151	PNSWAP	61	TX_MAIN_OUT
240	DUMMY	150	DSWAP[1]	60	DUMMY
239	DUMMY	149	DSWAP[0]	59	TVDD
238	DUMMY	148	FRM	58	VGL
237	VGHO	147	PLL_REF	57	VGL
236	VGHO	146	GPIO[1]	56	DUMMY
235	TX_AUX_OUT	145	HID_MODE_EN	55	TVSS
234	TVSS_ADC	144	LEDPWM	54	AVEE
233	TVDD_ADC	143	FTE1	53	TEST_TP[1]
232	DUMMY	142	FTE	52	TEST_TP[0]
231	VGLO	141	IM[0]	51	VCOM_OPT_L
230	VGLO	140	VDD_TP	50	VCOM
229	DUMMY	139	VDDI	49	VCOM
228	AVSS	138	VSS	48	VCOM_PASS_L
227	VMOD_GATE2	137	EN_DRIC_INI	47	VCOM_PASS_L
226	VMOD_GATE1	136	IM[1]	46	AVDD
225	VCOM_PASS_R	135	ENB_MULTI_DROP	45	AVDD
224	VCOM_PASS_R	134	RTERM	44	AVSS
223	AVDD	133	EN_RTERM	43	DUMMY
222	VCOM	132	EN_EXT_VCOM	42	VGHO
221	VCOM	131	EN_PWR_SEL	41	VGHO
220	VCOM_OPT_R	130	EN_EXT_HV	40	TVDD_ADC
219	TEST_TP[3]	129	EN_EXT_VDD	39	TVSS_ADC
218	TEST_TP[2]	128	AVEE_R	38	TX_AUX_OUT
217	AVEE	127	AVEE_R	37	DUMMY
216	TX_MAIN_OUT	126	VCOM_FB	36	VGH2
215	TEST[10]	125	VCOM	35	DUMMY
214	TX_MAIN_IN	124	VCOM	34	VGLO
213	TX_MAIN_IN	123	AVEE	33	VGLO
212	DUMMY	122	AVSS	32	SYNC_TP_DATA_L[7]
211	DUMMY	121	AVDD	31	SYNC_TP_DATA_L[6]
210	TVDD	120	VDD	30	SYNC_TP_DATA_L[5]
209	TVDD	119	VSS	29	SYNC_TP_DATA_L[4]
208	DUMMY	118	VDDI	28	SYNC_TP_DATA_L[3]
207	DUMMY	117	VP_HSSI	27	SYNC_TP_DATA_L[2]
206	DUMMY	116	VP_HSSI	26	SYNC_TP_DATA_L[1]
205	TVSS	115	VG_HSSI	25	SYNC_TP_DATA_L[0]
204	TVSS	114	HSSI_D3_N	24	SYNC_TP_CTRL_L[7]
203	DUMMY	113	VG_HSSI	23	SYNC_TP_CTRL_L[6]
202	VGL	112	VG_HSSI	22	SYNC_TP_CTRL_L[5]
201	VGL	111	HSSI_D3_P	21	SYNC_TP_CTRL_L[4]
200	C31N	110	VG_HSSI	20	SYNC_TP_CTRL_L[3]
199	C31P	109	HSSI_D0_N	19	SYNC_TP_CTRL_L[2]
198	AVEE	108	VG_HSSI	18	SYNC_TP_CTRL_L[1]
197	CVSS	107	VG_HSSI	17	SYNC_TP_CTRL_L[0]
196	AVDD	106	HSSI_D0_P	16	VDDI
195	C41N	105	VG_HSSI	15	VSS
194	C41P	104	HSSI_CLK_N	14	SYNC_DATA_L[7]
193	VCIP	103	VG_HSSI	13	SYNC_DATA_L[6]
192	VDD_TP	102	VG_HSSI	12	SYNC_DATA_L[5]
191	VDDI	101	HSSI_CLK_P	11	SYNC_DATA_L[4]
190	VSS	100	VG_HSSI	10	SYNC_DATA_L[3]
189	VDD	99	HSSI_D1_N	9	SYNC_DATA_L[2]
188	GVDDN	98	VG_HSSI	8	SYNC_DATA_L[1]
187	GVDDP	97	VG_HSSI	7	SYNC_DATA_L[0]
186	AVSS	96	HSSI_D1_P	6	DUMMY
185	AVDD	95	VG_HSSI	5	DUMMY
184	AVEE_DC	94	HSSI_D2_N	4	DUMMY
183	AVSS_DC	93	VG_HSSI	3	GND
182	AVDD_DC	92	VG_HSSI	2	DUMMY
181	VDDI	91	HSSI_D2_P	1	DUMMY

Product Specification

1-3-2. FPC Pin Assignment

Slave

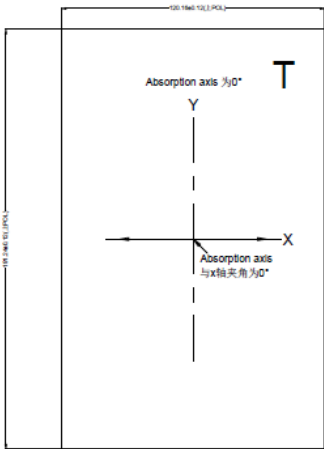
Please pay attention that IC bump down(TFT glass up and C/F glass down)

270	DUMMY	180	VDD	90	VG_HSSI
269	DUMMY	179	VDD_TP	89	VP_HSSI
268	GND	178	VDD_TP_PLL	88	VP_HSSI
267	DUMMY	177	VSS	87	VG_HSSI
266	DUMMY	176	PLL_XCK_IN	86	VDD
265	SYNC_DATA_R[0]	175	PLL_XCK_OUT	85	VSS
264	SYNC_DATA_R[1]	174	TCH_ATTEN	84	VDDI
263	SYNC_DATA_R[2]	173	TCH_IO_SDA	83	VDD_TP
262	SYNC_DATA_R[3]	172	TCH_IO_SCL	82	AVEE
261	SYNC_DATA_R[4]	171	FLASH_SPI_MISO	81	AVSS
260	SYNC_DATA_R[5]	170	FLASH_SPI_MOSI	80	VCIP
259	SYNC_DATA_R[6]	169	FLASH_SPI_SCK	79	AVDD
258	SYNC_DATA_R[7]	168	FLASH_SPI_CSN	78	CVSS
257	DUMMY	167	TCH_IO_SCK	77	C22N
256	VSS	166	TCH_IO_MOSI	76	C22P
255	VSS	165	TCH_IO_CSN	75	VGH
254	VDDI	164	TCH_IO_MISO	74	VGH
253	VDDI	163	TP_OPT	73	CVSS
252	SYNC_TP_CTRL_R[0]	162	GPIO[0]	72	AVEE
251	SYNC_TP_CTRL_R[1]	161	TP_MS	71	AVDD
250	SYNC_TP_CTRL_R[2]	160	VDDI_DC	70	C21N
249	SYNC_TP_CTRL_R[3]	159	VSS	69	C21P
248	SYNC_TP_CTRL_R[4]	158	VDDI	68	DUMMY
247	SYNC_TP_CTRL_R[5]	157	DUMMY	67	AVDD
246	SYNC_TP_CTRL_R[6]	156	RESX	66	DUMMY
245	SYNC_TP_CTRL_R[7]	155	ENB_CASC	65	DUMMY
244	SYNC_TP_DATA_R[0]	154	LCD_LR	64	VDD
243	SYNC_TP_DATA_R[1]	153	VDDIDUM[0]	63	DUMMY
242	SYNC_TP_DATA_R[2]	152	TP_EN	62	TX_MAIN_IN
241	SYNC_TP_DATA_R[3]	151	PNSWAP	61	TX_MAIN_OUT
240	SYNC_TP_DATA_R[4]	150	DSWAP[1]	60	DUMMY
239	SYNC_TP_DATA_R[5]	149	DSWAP[0]	59	TVDD
238	SYNC_TP_DATA_R[6]	148	FRM	58	VGL
237	SYNC_TP_DATA_R[7]	147	PLL_REF	57	VGL
236	DUMMY	146	GPIO[1]	56	DUMMY
235	VGHO	145	HID_MODE_EN	55	TVSS
234	VGHO	144	LEDPWM	54	AVEE
233	TX_AUX_OUT	143	FTE1	53	TEST_TP[1]
232	TVSS_ADC	142	FTE	52	TEST_TP[0]
231	TVDD_ADC	141	IM[0]	51	VCOM_OPT_L
230	DUMMY	140	VDD_TP	50	VCOM
229	VGLO	139	VDDI	49	VCOM
228	VGLO	138	VSS	48	VCOM_PASS_L
227	DUMMY	137	EN_DRIC_INI	47	VCOM_PASS_L
226	AVSS	136	IM[1]	46	AVDD
225	VMOD_GATE2	135	ENB_MULTI_DROP	45	AVDD
224	VMOD_GATE1	134	RTERM	44	AVSS
223	VCOM_PASS_R	133	EN_RTERM	43	DUMMY
222	VCOM_PASS_R	132	EN_EXT_VCOM	42	VGHO
221	AVDD	131	EN_PWR_SEL	41	VGHO
220	VCOM	130	EN_EXT_HV	40	TVDD_ADC
219	VCOM	129	EN_EXT_VDD	39	TVSS_ADC
218	VCOM_OPT_R	128	AVEE_R	38	DUMMY
217	TEST_TP[3]	127	AVEE_R	37	TX_AUX_OUT
216	TEST_TP[2]	126	VCOM_FB	36	VGH2
215	AVEE	125	VCOM	35	DUMMY
214	TX_MAIN_OUT	124	VCOM	34	VGLO
213	TEST[10]	123	AVEE	33	VGLO
212	TX_MAIN_IN	122	AVSS	32	DUMMY
211	TX_MAIN_IN	121	AVDD	31	DUMMY
210	TVDD	120	VDD	30	DUMMY
209	TVDD	119	VSS	29	DUMMY
208	DUMMY	118	VDDI	28	DUMMY
207	DUMMY	117	VP_HSSI	27	DUMMY
206	DUMMY	116	VP_HSSI	26	DUMMY
205	TVSS	115	VG_HSSI	25	DUMMY
204	TVSS	114	HSSI_D3_N	24	DUMMY
203	DUMMY	113	VG_HSSI	23	DUMMY
202	VGL	112	VG_HSSI	22	DUMMY
201	VGL	111	HSSI_D3_P	21	DUMMY
200	C31N	110	VG_HSSI	20	DUMMY
199	C31P	109	HSSI_D0_N	19	DUMMY
198	AVEE	108	VG_HSSI	18	DUMMY
197	CVSS	107	VG_HSSI	17	DUMMY
196	AVDD	106	HSSI_D0_P	16	VDDI
195	C41N	105	VG_HSSI	15	VSS
194	C41P	104	HSSI_CLK_N	14	FOGTEST2
193	VCIP	103	VG_HSSI	13	FOGTEST1
192	VDD_TP	102	VG_HSSI	12	TPDE
191	VDDI	101	HSSI_CLK_P	11	TPDO
190	VSS	100	VG_HSSI	10	CT6
189	VDD	99	HSSI_D1_N	9	CT5
188	GVDDN	98	VG_HSSI	8	CT4
187	GVDDP	97	VG_HSSI	7	CT3
186	AVSS	96	HSSI_D1_P	6	CT2
185	AVDD	95	VG_HSSI	5	CT1
184	AVEE_DC	94	HSSI_D2_N	4	DUMMY
183	AVSS_DC	93	VG_HSSI	3	GND
182	AVDD_DC	92	VG_HSSI	2	AG_GND
181	VDDI	91	HSSI_D2_P	1	DUMMY

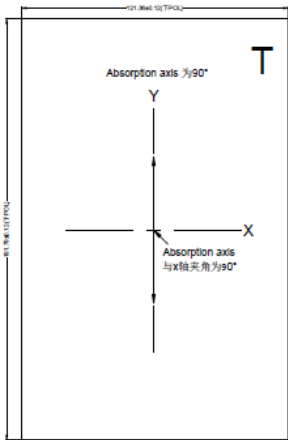
Product Specification

1-3-3. Pol General Spec

Mark 于保护膜之右上角，上红，下蓝



POL-CF
料号:3130502A30000T
型号: GC20 HC (AS)
厚度: 75 μm
保护膜朝上，离型膜向下



POL-TFT
料号:3130502A40000T
型号: GC2X APF
厚度: 78 μm
保护膜朝上，离型膜向下

Pol Mode Name				GC20 HC (AS)	GC2X APF
Products	outline	①width (Lx)	mm	120.16±0.12	121.36±0.12
		②length (Ly)	mm	191.24±0.12	191.79±0.12
	absorption angle(PL)		°	0.0 ±0.5	90.0±0.5
	APF Penetration angle(PL)		°	-	0.0±4.0
	right-angle		°	90.00 ± 0.08	90.00 ± 0.08
Thickness	Pol Total Thickness(with protect flim and release film)		um	206±20	169±20
	Pol Effective Thickness		um	75 ±20	78 ± 20
	Protect Film Thickness		um	93±5	53±5
	Release Film Thickness		um	38±5	38±5
	Glue Thickness		um	13±5	13±5

Product Specification	
-----------------------	--

1-4. Change List (新开案)

[illegible]

Remarks:

- 1.H/W change includes all materials, components, label etc.
- 2.Design change includes size, position etc.
- 3.S/W change includes initial code etc.

Please pay attention that this list is just a summary, an individual Design Change List which contain much more information is also needed

Month. Day. Year

Product Specification

2-1. Absolute Maximum Ratings

The following are maximum values which, if exceeded, may cause faulty operation or damage to the unit.

Table 2.1 Absolute Maximum Ratings

Parameter	Symbol	Values		Units	Notes
		Min	Max		
Supply voltage	VDDI	-0.3	+2.1	V	
Driver supply voltage	AVDD-AVSS	-0.3	+6.5	V	
Driver supply voltage	AVEE-AVSS	-6.5	+0.3	V	
Driver supply voltage	VGH-VGL	-0.3	32	V	
Operating temperature range	TOPR	-30	+75	°C	
Storage temperature range	TSTG	-40	+85	°C	
Logic input voltage range	VIN	-0.3	+VDDI+0.3V	V	
Logic output voltage range	VO	-0.3	+VDDI+0.3V	V	
Storage Humidity		5%	95%	%	
Storage Temperature		-30°C	70°C	°C	

Product Specification

3-1. ELECTRICAL CHARACTERISTICS

Table 3.1 Electrical Characteristics Of TFT-LCD Module

Parameter		Symbol	Values			Units	Notes
			Min	Typ	Max		
Logic Supply Voltage		IOVCC	1.83	1.93	2.03	V	Ta=-30~75℃, VGH-VGL must less than 32V
		Power ripple	/	120	280	mV	示波器测试Vpp电压值
		Power peak current	/	320	380	mA	示波器测试Ipp电流值
Positive source driver power (External input mode)		VSP	5.6	5.8	6.0	V	Ta=-30~75℃, VGH-VGL must less than 32V
		Power ripple	/	180	500	mV	示波器测试Vpp电压值
		Power peak current	/	250	300	mA	示波器测试Ipp电流值
		VSN	-6.0	-5.8	-5.6	V	Ta=-30~75℃, VGH-VGL must less than 32V
		Power ripple	/	220	500	mV	示波器测试Vpp电压值
		Power peak current	/	300	360	mA	示波器测试Ipp电流值
CTP Power Supply Voltage		VDDA	/	/	/	V	
Input Signal Voltage	Low Level	V _{IL}	VSS	/	0.2IOVCC	V	Applicable Pin : REST
	High Level	V _{IH}	0.8IOVCC	/	IOVCC	V	Applicable Pin : REST
Ouput Voltage	Low Level	V _{OH}	0.8IOVCC	/	IOVCC	V	Applicable Pin : BC I OUT = -0.1mA
	High Level	V _{OL}	VSS	/	0.2IOVCC	V	Applicable Pin : BC I OUT = +0.1mA

Notes:
The specified current and power consumption are under the conditions at **VSP=5.8V**, **VSN=-5.8V**,**IOVCC=1.93V**,
T = 25℃, and fv = **165** Hz, at white pattern

3-2. Logic Power Consumption

Parameter	Symbol	Values		Units	Notes
		Typ	Max		
Normal Mode	I _{IOVCC 1.93v}	125	150	mA	White Pattern with touch @165 Hz
	I _{vsp +5.8v}	24	31.5	mA	
	I _{vsn -5.8v}	18	26	mA	
	I _{IOVCC 1.93v-peak}	199	239	mA	Peak current
	I _{vsp +5.8v-peak}	173	208	mA	
	I _{vsn -5.8v-peak}	217	260	mA	
Sleep Mode	I _{IOVCC 1.93v}	/	2000	uA	Sleep in:2810 + LPWG
	I _{vsp +5.8v}	/	300	uA	
	I _{vsn -5.8v}	/	200	uA	

Product Specification

3-3. BACK LIGHT UNIT

3-3-1 The edge-lighting type of back light unit consists of 36 (9*4) LEDs which is connected in serial.

Table 2-3-1 Electrical Characteristics Of Back Light Unit

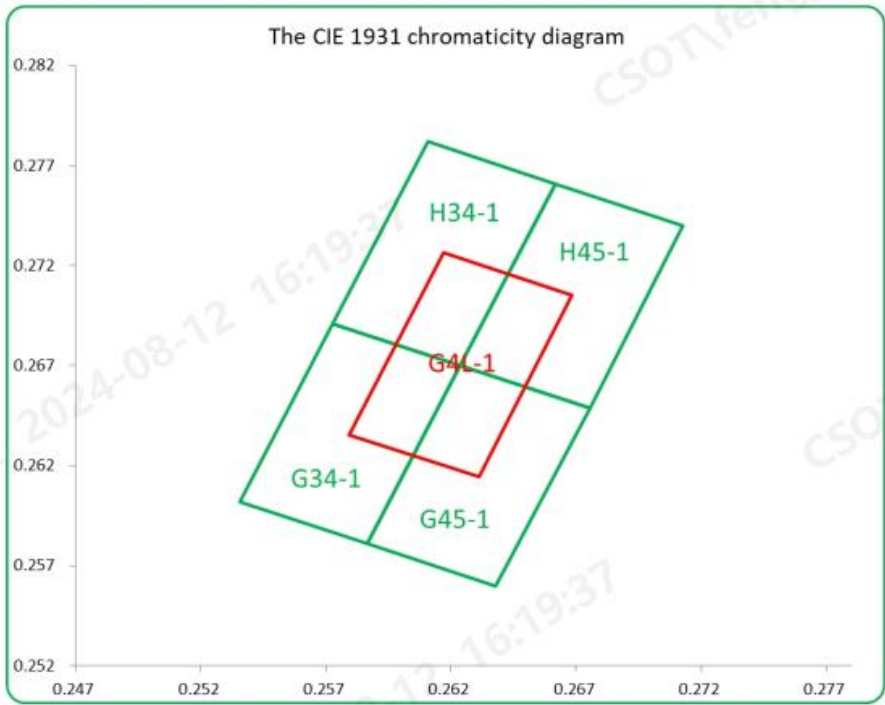
Parameter	Symbol	Values			Units	Notes
		Min	Typ.	Max		
LED Current	I _{LED}	/	23	/	mA	750nits
LED Forward Voltage	V _{LED}	2.7	2.8	2.9	V	

3-3-2 LED Rank

Luminous Flux Rank Chart

Bin code	Min (mcd)	Max (mcd)	Min (lm)	Max (lm)
37	3050	3150	8.50	8.75
38	3150	3250	8.75	9.00

Color Rank Chart



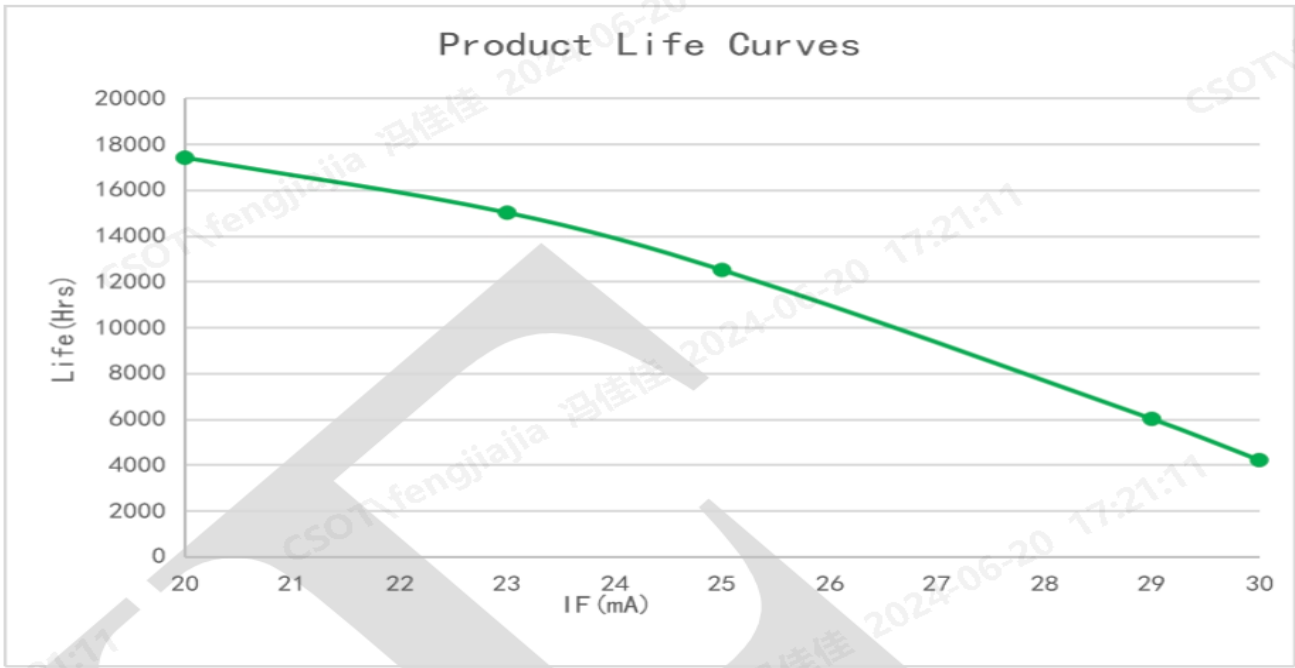
Product Specification

3-3-3 LED current&TPM light

LED Current(mA)	TPM light(nits)
15.6	500
18.8	600
22.0	700
23.0	750
25.2	800
29.2	900

3-3-4 The minimum life curves of different current drivers are as follows:

IF (mA)	20	23	25	29	30
tmin (Hrs)	17400	15000	12500	6000	4200



Product Specification

3-4. LCD INTERFACE CONNECTIONS

Interface Connector: Model name(Supplier)

Table 3.4 LCD Connector Pin Configuration

Pin No.	Symbol	I/O	Description
1	GND	P	GND
2	C2_S	I	MIPI input
3	B2_S	I	MIPI input
4	A2_S	I	MIPI input
5	GND	P	GND
6	C1_S	I	MIPI input
7	B1_S	I	MIPI input
8	A1_S	I/O	MIPI I/O
9	GND	P	GND
10	C0_S	I/O	MIPI I/O
11	B0_S	I	MIPI input
12	A0_S	I	MIPI input
13	GND	P	GND
14	GND	P	GND
15	LED_A	P	LED+
16	LED_A	p	LED+
17	DUMMY	/	/
18	DUMMY	/	/
19	LED_K1	P	LED-
20	LED_K2	P	LED-
21	LED_K3	P	LED-
22	LED_K4	p	LED-
23	DUMMY	/	/
24	DUMMY	/	/
25	RESET	Digital input	Reset device
26	TP_INT	Digital output	Touch interrupt pin
27	TP_SPI_SCK	Digital input	Flash SPI interface: clock input
28	TP_SPI_MISO	Digital output	Flash SPI interface: data output
29	TP_SPI_CS	Digital input	Flash SPI interface: chip select
30	TP_SPI_MOSI	Digital input	Flash SPI interface: data iutput

Product Specification

3-4. LCD INTERFACE CONNECTIONS

Interface Connector: Model name(Supplier)

Table 3.4 LCD Connector Pin Configuration

Pin No.	Symbol	I/O	Description
31	PWM	Digital output	Connect external LED driver of panel backlight control (CABC)
32	FTE_S	Digital output	Slave IC Frame head pulse signal. For debug
33	FTE_M	Digital output	Master IC Frame head pulse signal. For debug
34	ID1	/	Panel provider select
35	ID0	/	Panel provider select
36	IM0(DUMMY)	/	CSOT IM0
37	VDDI	P	Power supply to the I/O
38	VDDI	P	Power supply to the I/O
39	VDDI	P	Power supply to the I/O
40	DUMMY	/	/
41	VSN	P	Negative input analog power for driver IC use
42	VSN	P	Negative input analog power for driver IC use
43	VSN	P	Negative input analog power for driver IC use
44	DUMMY	/	/
45	VSP	P	Positive input analog power for driver IC use
46	VSP	P	Positive input analog power for driver IC use
47	VSP	P	Positive input analog power for driver IC use
48	DUMMY	/	/
49	GND	P	GND
50	C2_M	I	MIPI input
51	B2_M	I	MIPI input
52	A2_M	I	MIPI input
53	GND	P	GND
54	C1_M	I	MIPI input
55	B1_M	I	MIPI input
56	A1_M	I/O	MIPI I/O
57	GND	P	GND
58	C0_M	I/O	MIPI I/O
59	B0_M	I	MIPI input
60	A0_M	I	MIPI input
61	GND	P	GND

Product Specification

3-5. SIGNAL TIMING SPECIFICATIONS(reference for CSOT AP)

ITEM			SYMBOL	Timing	UNIT
LCD	Frame Rate		-	165	Hz
Timing	DCLK	Frequency	fCLK	570.9	MHz
		Period	Tclk	1.75	ns
	Horizontal	Horizontal total time	tHP	1756	t _{CLK}
		Horizontal Active time	tHadr	1600	t _{CLK}
		Horizontal Pulse Width	tHsync	10	t _{CLK}
		Horizontal Back Porch	tHBP	72	t _{CLK}
		Horizontal Front Porch	tHFP	74	t _{CLK}
	Vertical	Vertical total time	tvp	2850	t _H
		Vertical Active time	tVadr	2560	t _H
		Vertical Pulse Width	tVsync	4	t _H
		Vertical Back Porch	tVBP	256	t _H
		Vertical Front Porch	tVFP	30	t _H
Differential Swing			VDswing		mV
Bit Rate			TX SPD(MBPS)	1100	Mbps
Pixel Fomat				pixel	Data bit/pixel
Lane				2 port 6 trios	Lane

Product Specification

3-6. MIPI Data&CLK Line Impedance Test Result (TBD)
MIPI阻抗：45±5Ω



MIPI阻抗数据-rawdata

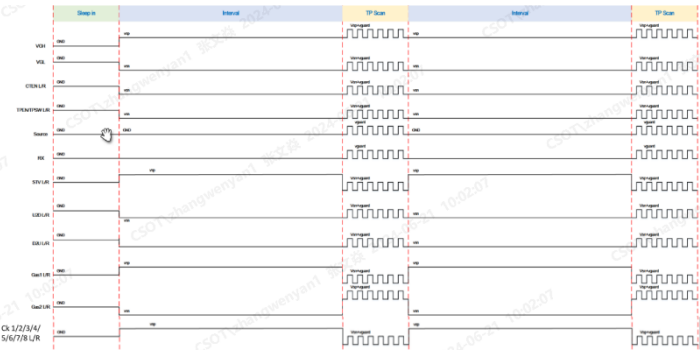
Product Specification

3-7. Power On/Off Sequence
Power on/off timing

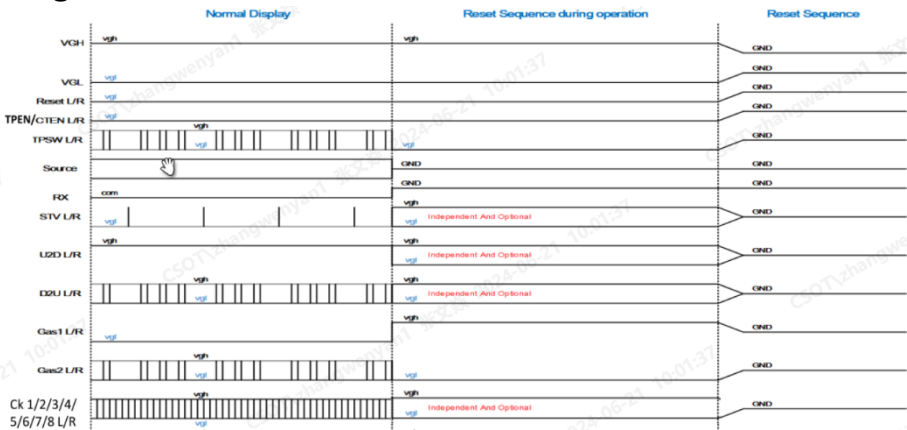
5.9.1.2 Three power (VDDI/ AVDD/ AVEE) – No Flash Mode



LPWG timing



APO timing



Product Specification

3-7. Power On/Off Sequence
Power on/off timing

Item	Symbol	Condition	Min	Typ	Max	Unit
AVDD - AVEE delay time (10% to 10%)	tPON1	Power On	0	-	-	us
AVDD - AVEE delay time (90% to 90%)	tPON2	Power On	0	-	-	us
VDDI On to AVDD On time (90% to 10%)	tsVSP	Power On	1	-	-	ms
All Power On to RESX Hi-level time (90% to VIH)	tRW	Power On	10	-	-	ms
RST Hi-level (VIH) to I2C Boot reset time	tRT2BRST	Power On	10	-	-	ms
RST Hi-level (VIH) to 1st Command time	tRT1(without OTP in flash mode)	Power On	10	-	-	ms
RST Hi-level (VIH) to 1st Command time	tRT1(with OTP in flash mode)	Power On	90	-	-	ms
(MIPI goes to LP11 state) to RST goes to Low-level (VIL)	tRT2	Power On	-	0	-	ms
SLPOUT Sequence Request time	tsLPOUT	Power On	100	-	-	ms
MIPI Video Stream On time to BL On time	tBLON	Power On	40	-	-	ms
AVDD - AVEE delay time (10% to 10%)	tPOFF1	Power Off	0	-	-	us
AVDD - AVEE delay time (90% to 90%)	tPOFF2	Power Off	0	-	-	us
AVDD Off to VDDI Off time (10% to 90%)	thVSP	Power Off	0	-	-	us
RST Low to VDDI Off time (VIH to 90%)	tROFF	Power Off	0	-	-	us
BL Off to Display Off Command time	tBLOFF	Power Off	0	-	-	us
SLPIN Sequence Request time	tsLPIN	Power Off	60	-	-	ms
VDD regulator turn off time	tVDDOFF	Suspend	20	-	80	ms
MIPI Video Stream Off time to DISPLAY OFF command	tsOFF1	Power Off	0	-	-	us
RST goes to Low-level (VIH) to MIPI goes to LP00 state	tsOFF2	Power Off	0	-	-	ms
I2C Power Down to DISPLAY OFF command	tPD2SOFF	Power Off	50	-	-	ms
SPI Wake Up Gesture to DISPLAY OFF command	tWKG2SOFF	Power Off	50	-	-	ms

Product Specification

3-8. Software Flow

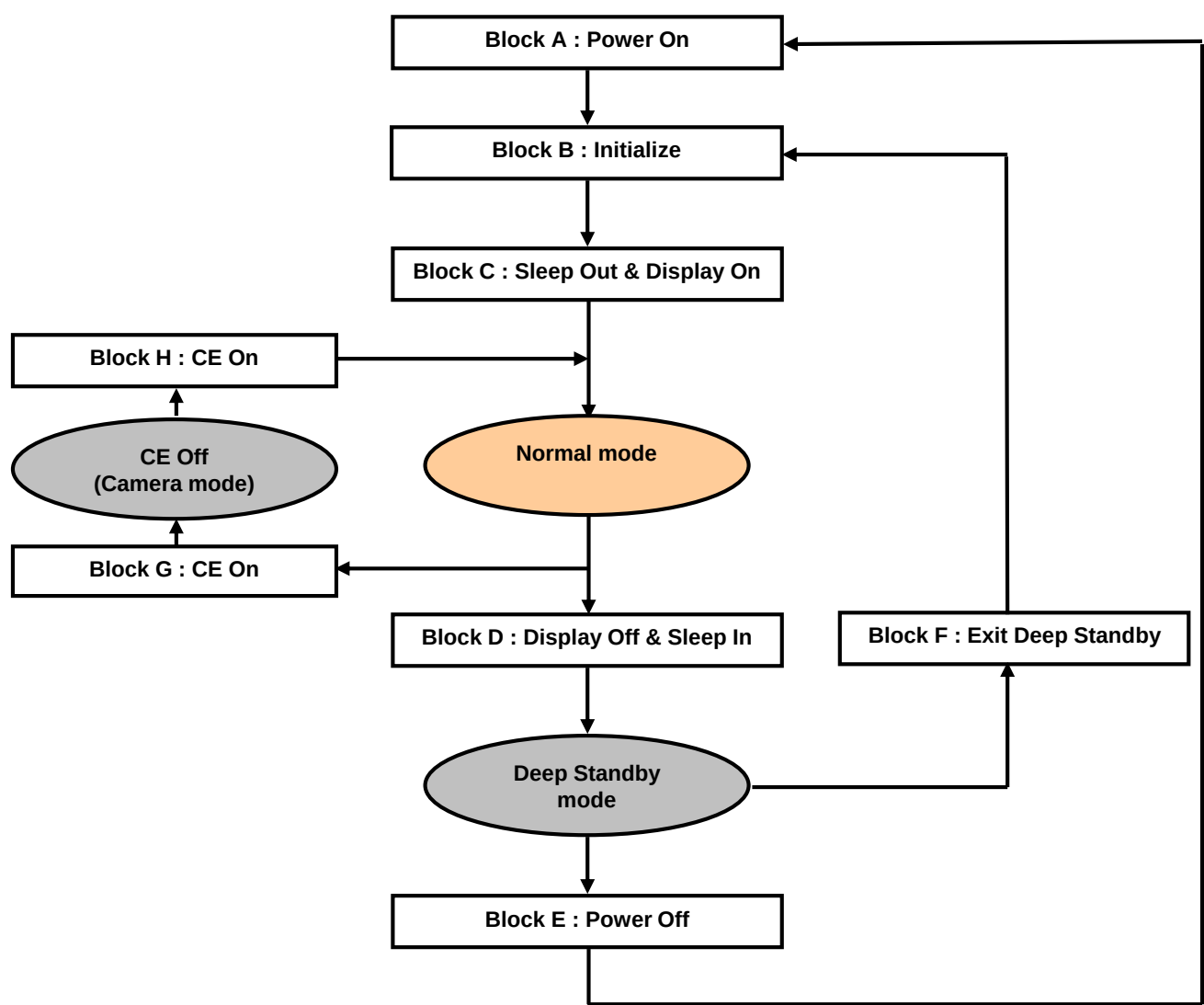


Fig 3.7 Software Flowchart

Product Specification

3-9. Initial Code

3-9-1: Power on

Step	Command Parameter	Command name	Operation
1	Base on AP CMD		
2			
3			
4			
5			
6			

3-9-2: Initialize

Step	Command Parameter	Command name	Operation
1	39 01 00 00 00 00 02 FF D0 39 01 00 00 00 00 02 FB 01 39 01 00 00 00 00 02 00 30 39 01 00 00 00 00 02 09 EE 39 01 00 00 00 00 02 1C 99 39 01 00 00 00 00 02 1D 09 39 01 00 00 00 00 02 FF F0 39 01 00 00 00 00 02 FB 01 39 01 00 00 00 00 02 24 02 39 01 00 00 00 00 02 FF 25 39 01 00 00 00 00 02 FB 01 39 01 00 00 00 00 02 19 FF 39 01 00 00 00 00 02 1B 1B 39 01 00 00 00 00 02 1C F0 39 01 00 00 00 00 02 F2 00 39 01 00 00 00 00 02 F3 00 39 01 00 00 00 00 02 FF 10 39 01 00 00 00 00 02 FB 01 39 01 00 00 00 00 02 35 00 39 01 00 00 00 00 03 51 0F FF 39 01 00 00 00 00 02 53 2C 39 01 00 00 00 00 02 55 00 39 01 00 00 00 00 02 90 03 39 01 00 00 00 00 11 91 89 28 00 14 D2 00 02 B2 02 2B 00 0B 05 7A 03 68 39 01 00 00 00 00 03 92 10 E0 ### 刷新频率切换code ###DP-165Hz --> 0xB2=0x80 / 0xB3=0x20 ###DP-144Hz --> 0xB2=0x80 / 0xB3=0x12 ###DP-120/60/30Hz --> 0xB2=0x91 / 0xB3=0x01 ###DP-90Hz --> 0xB2=0x80 / 0xB3=0x13 ###165Hz 39 01 00 00 00 00 02 B2 80 39 01 00 00 00 00 02 B3 20 39 01 00 00 00 00 02 BB 13 39 01 00 00 00 00 07 3B 03 04 1E 04 04 01 39 01 00 00 00 00 02 11 00 39 01 00 00 00 00 02 29 00		

Product Specification

3-9-3: Sleep out & Display on

Step	Command Parameter	Command name	Operation
1	11 29	Specific ways are based on AP CMD	

3-9-4: Display off & Sleep in

Step	Command Parameter	Command name	Operation
1	28 10	Specific ways are based on AP CMD	

Product Specification

3-9-5: Display off & Sleep in

Step	Command Parameter	Command name	Operation
1		In previous page	
2			

3-9-6: Power off

Step	Command Parameter	Command name	Operation
1	Base on AP CMD		
2			

3-9-7: Exit deep standby

Step	Command Parameter	Command name	Operation
1	(enter deep standby)NOVA ps tool, send CMD 13→28 10		
2	(exit deep standby)only AP can do		
3			

3-9-8: CE off

Step	Command Parameter	Command name	Operation
1	FF10		
	0X55 0X00		

3-9-9: CE on

Step	Command Parameter	Command name	Operation
1	FF10		
	0X55 0XA0		

Product Specification

3-9-10: CABC off (NO USE)

Step	Command Parameter	Command name	Operation
1	FF10	\	\
	0X55 0X00		\

3-9-11: CABC on (NO USE)

Step	Command Parameter	Command name	Operation
1	FF10	\	\
	0X55 0XA0		\

3-9-12: ACO off (NO USE)

Step	Command Parameter	Command name	Operation
1	NO USE	\	\
			\

3-9-13: ACO on (NO USE)

Step	Command Parameter	Command name	Operation
1	NO USE	\	\
			\

3-9-14: SRE off (NO USE)

Step	Command Parameter	Command name	Operation
1	FF10	\	\
	0X55 0X00		\

3-9-15: SRE on (NO USE)

Step	Command Parameter	Command name	Operation
1	FF10	\	\
	0X55 0XA1		\

Product Specification**3-10. IC General Spec and Size****1. General Description****1.1 Purpose of this Document**

This document has been created to provide complete reference specifications for the NT36523. IC designers, testing engineers and application engineers should refer to these specifications for circuits design, quality/performance control, and IC applications for customer.

1.2 General Description

The NT36523 device is a single or two chip touch with RAM-less display driver integration IC for a-Si, IGZO, LTPS GOA TFT LCD that incorporates gate drivers, a timing controller with glass interface level-shifters, a VCOM driver and a glass power supply circuit.

The NT36523 can support MIPI interface. The source resolution only can be adjusted 800RGB and the gate resolution only can be set to 2560 lines. About the detailed resolution setting, please refer to NT36523 Application Note.

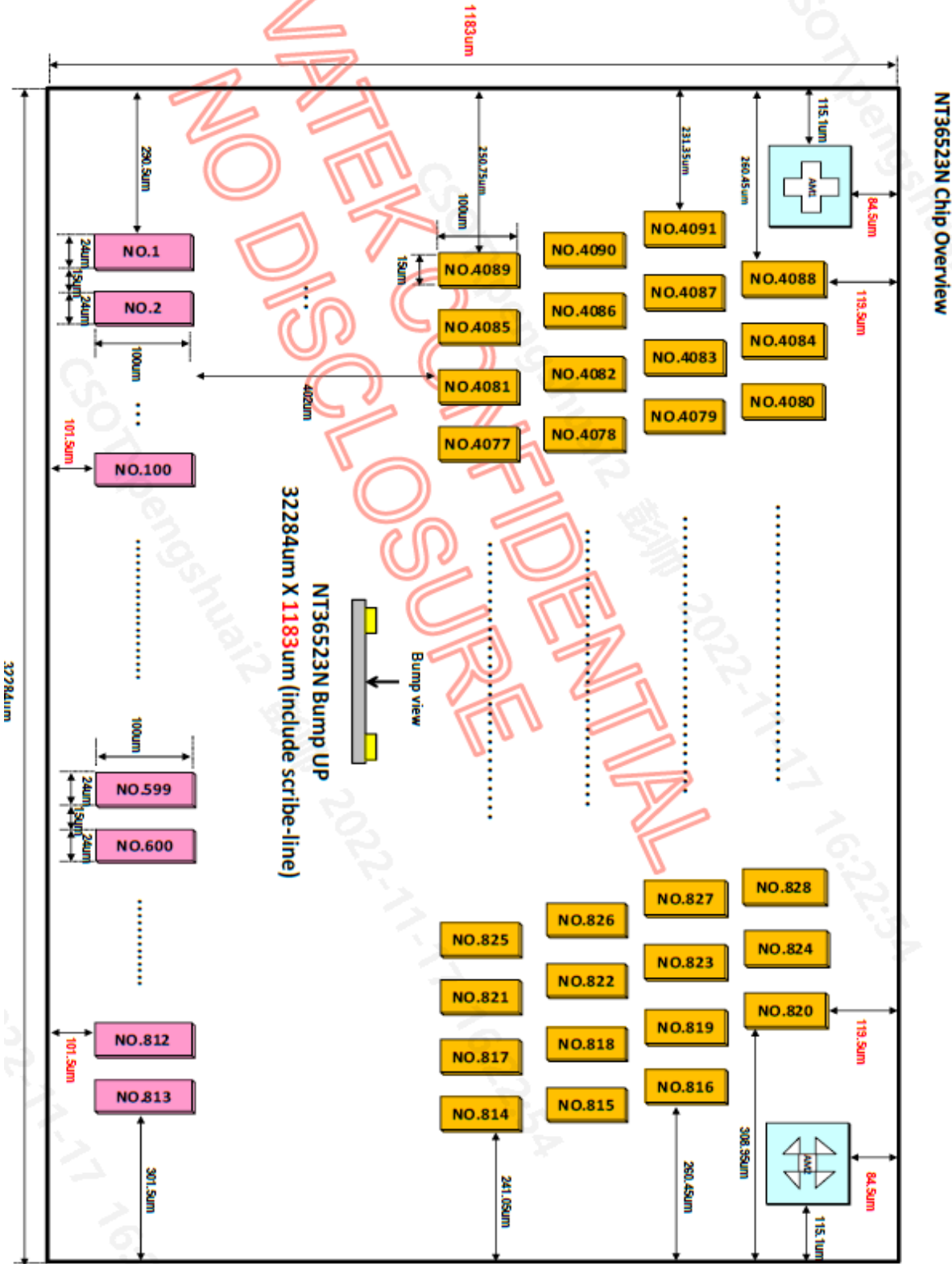
The NT36523 is also able to make gamma correction settings separately for RGB dots to allow benign adjustments to panel characteristics, resulting in higher display qualities. The IC includes internal boosters that generate the LCD driving voltage, breeder resistance and voltage follower circuit for the LCD driver. A deep standby mode is also supported for lower power consumption.

The NT36523 also supports CABC function for the backlight control. It's able to reduce the total power consumption of display module significantly.

This LSI is suitable for small or medium-sized tablet solutions requiring long-term driving capabilities.

Product Specification

3-10. IC General Spec and Size



Product Specification

3-11. Power Consumption

	Parameter	Symbol	Typ	Unit	Remarks
CABC off	Logic Power@165 Hz	L0	434.1	Mw	
		L32	504.5	Mw	
		L64	512.1	Mw	
		L96	516.0	Mw	
		L127	521.7	Mw	
		L160	510.1	Mw	
		L192	509.9	Mw	
		L224	519.0	Mw	
		L256	481.7	Mw	
		R255	449.6	Mw	
		G255	453.1	Mw	
		B255	453.7	Mw	
		8 color bar	461.0	Mw	
		0-255 Gray Transition	584.7	Mw	
		Lenovo UI	641.1	Mw	Lenovo will send pic.
		Icon Interface	/	Mw	
	BLU	25% on	0.498248	w	
		50% on	1.046938	w	
		75% on	1.652798	w	
		100% on	2.307732	w	

	Symbol	Logic	BLU	Remarks
CABC on (No CABC USE)	L0	\	\	\
	L32	\	\	\
	L64	\	\	\
	L96	\	\	\
	L127	\	\	\
	L160	\	\	\
	L192	\	\	\
	L224	\	\	\
	L256	\	\	\
	R255	\	\	\
	G255	\	\	\
	B255	\	\	\
	8 color bar	\	\	\
	0-255 Gray Transition	\	\	\
	Lenovo UI	\	\	Lenovo will send pic.
	Icon Interface	\	\	

If there is more than one mode for CABC function, please fill out all data

Product Specification

3-12. Power Consumption-Other Frame Rate

Parameter	Frame Rate	Symbol	Typ	Unit	Remarks
Logic Power	165Hz+Hand	White	483	Mw	
		Beauty	737	Mw	
		Subpixel	1923	Mw	
	144Hz+Hand	White	449	Mw	
		Beauty	678	Mw	
		Subpixel	1378	Mw	
	120Hz+Pen	White	497	Mw	
		Beauty	696	Mw	
		Subpixel	1311	Mw	
	90Hz+Hand	White	415	Mw	
		Beauty	568	Mw	
		Subpixel	987	Mw	
	60Hz+Pen	White	464		
		Beauty	561		
		Subpixel	875		
	30Hz+Pen	White	456		
		Beauty	527		
		Subpixel	672		

Product Specification

4. OPTICAL CHARACTERISTICS

4-1. Optical Characteristics – Backlight 100%

Parameter	Symbol	Condition	Min	Typ	Max	Unit	Remarks
Viewing Angle	Θ12	CR >10	80	-	-	-	Note 1
	Θ6		80	-	-	-	Note 1
	Θ9		80	-	-	-	Note 1
	Θ3		80	-	-	-	Note 1
Contrast Ratio	CR	Optimal	1000	1500	-	-	Note 1,4
Brightness	Lv	23mA	640	750	-	Cd/m ²	Note 1
		15.6mA	425	500	-		
Brightness Uniformity	Y	Optimal	80	85	-	%	Note 1.7(13P)
	Y	Optimal	65	70	-	%	Note 1.7(160P) 暫定
Color uniformity			-	-	0.010	-	Note 1.7(13P)
CCT Range			-	-	1000	K	Note 1.7(13P)
CCT			6100	7500	9800	K	
Gamma			L20~L224:2.2±0.2 L12~L20&L224~L240:2.2±0.4 L0~L12&L240~L255: 不卡控			-	
Low blue light		Area ratio of (415~455nm) / (400~500nm)	-	-	50%	%	
Flicker			-	-	-30	dB	Note 1,2
Crosstalk			-	-	2	%	Note 1,3
Response time	T _f + T _r	Θ = 0 ° Ta = 25 °C	-	17	20	ms	Note 1,6
	GTG		-	-	24	ms	Note 1,6
Color Gamut	DCI-P3	-	91	96	-	%	Note 1
	NTSC	-	91	96	-	%	
White Chromaticity	x	CIE 1931	0.279	0.299	0.319	-	Note 1
	y		0.295	0.315	0.335	-	Note 1
Red Chromaticity	x	CIE 1931	0.661	0.681	0.701	-	Note 1
	y		0.296	0.316	0.336	-	Note 1
Green Chromaticity	x	CIE 1931	0.236	0.256	0.276	-	Note 1
	y		0.660	0.680	0.700	-	Note 1
Blue Chromaticity	x	CIE 1931	0.119	0.139	0.159	-	Note 1
	y		0.044	0.064	0.084	-	Note 1

4-2. Cell&BLU Optical Characteristics

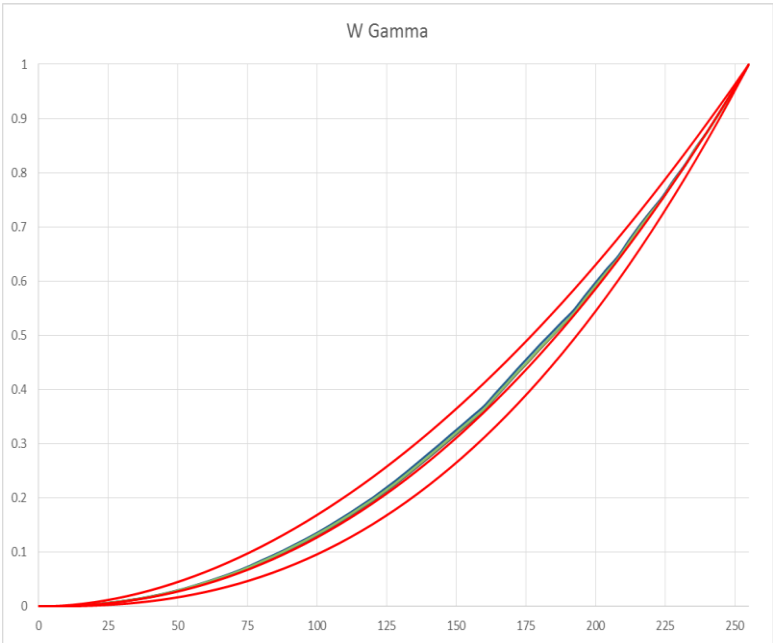
Parameter	Typ	Unit	Remarks
Aperture Ratio	54.25	%	
Upper Pol Trans.	43.5	%	
Lower Pol Trans.	41.5	%	
Panel Trans.	4.27	%	w/o APF
Panel Trans.	5.98	%	with APF
BLU Luminance	12550	Cd/m ²	Center
BLU Luminance Uniformity	83min	%	13点,83%min

Product Specification

4-3. Gamma/Color Coordinate Uniformity/CE Function/CABC Function

4.3-1 Gamma Curve

Request: W, 0-255 gray scale, step 4 gray scale, refer to Lenovo TFT-LCD test plan V3.5 (tablet).

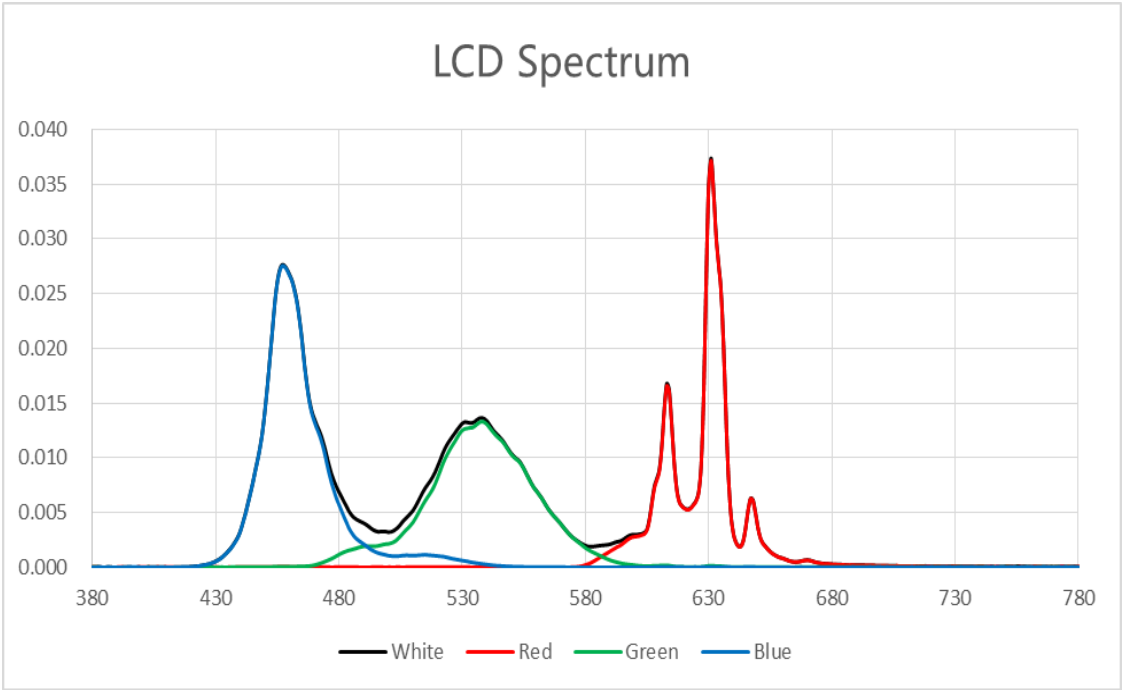


4.3-2 CABC function(on and off)

Request: Pictures data comparison(Lenovo UI, icon pattern, L255, L192, L127, L64, L0, some colorful pattern),
Movies comparison(three segments with different frames details, lighter, light+dark, darker)

Product Specification

4-4. LCD Spectrum and BLU Spectrum
Center Point



Product Specification

[Note 1] Optical Test Equipment Setup

The optical characteristics should be measured in dark room. After 5 minutes operation, the optical properties are measured at the center point of the LCD screen. (Viewing angle is measured by PR740 or CS2000 /Height :500mm,

Response time is measured by Optoscope, other items are measured by CS2000/PR740/ Field of view: 1°/Height: 500mm)

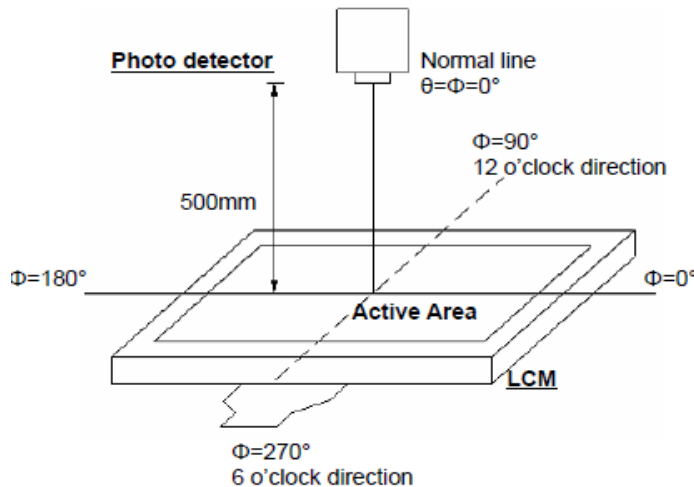
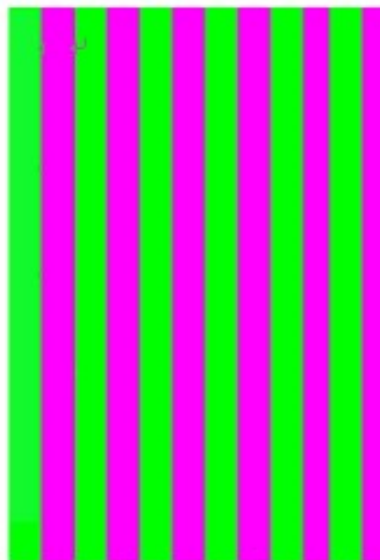


Fig 4.1. Optical Characteristic Measurement Equipment and Method

[Note 2] Flicker

The flicker level should be measured with horizontal Magenta 128 /Green 128 stripes. The flicker is essentially a ratio of the powers in the frequency spectrum at 30 Hz (P_X) and 0 Hz (P_0 , DC level).

$$F = 20 \text{ Log } (P_X / P_0)$$



Product Specification

[Note 3] Crosstalk

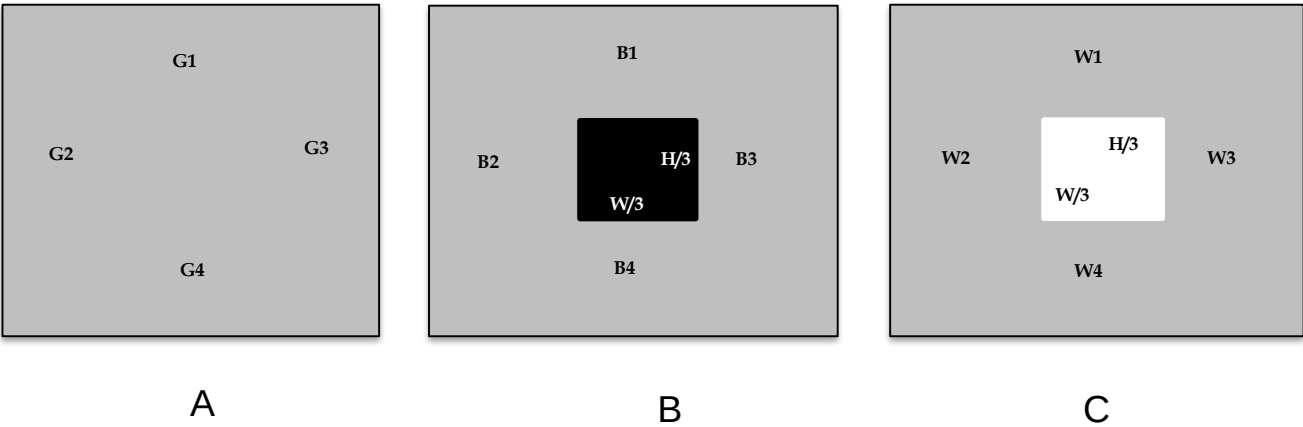


Fig 4.2. Crosstalk measurement points & pattern

- A: Luminance for P1 ~ P4 with all 127gray pixels
- B: Luminance for P1 ~ P4 with 127gray pixels when the black box is applied
- C: Luminance for P1 ~ P4 with 127gray pixels when the white box is applied

Black Crosstalk [%] = Maximum $\left[\text{Absolute} \left(\frac{B - A}{A} \right) \right]$

White Crosstalk [%] = Maximum $\left[\text{Absolute} \left(\frac{C - A}{A} \right) \right]$

Product Specification

[Note 4]

Contrast Ratio is defined as follows ;

Contrast Ratio(CR) =
$$\frac{\text{Photo detector output with LCD being "White"}}{\text{Photo detector output with LCD being "Black"}}$$

[Note 5]

Viewing Angle Range is defined as follows;

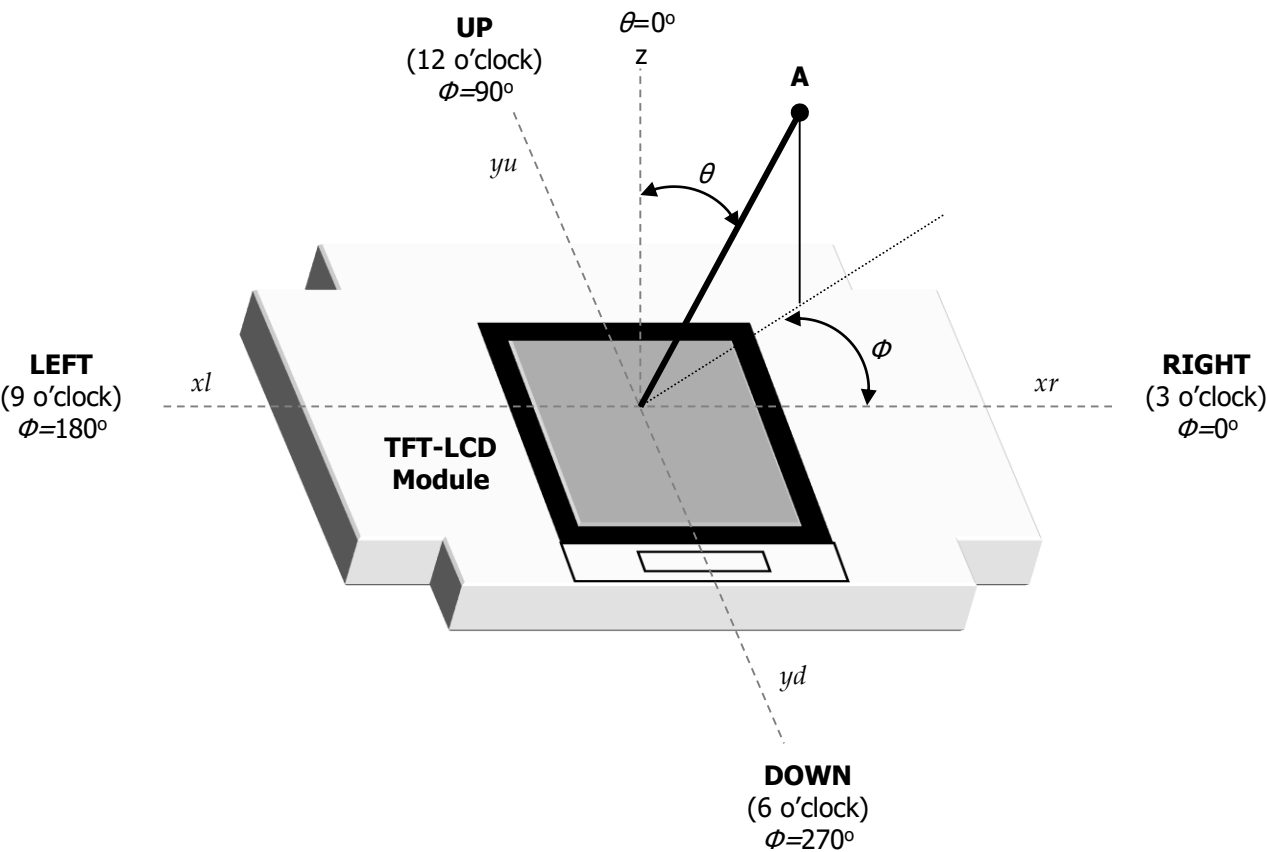
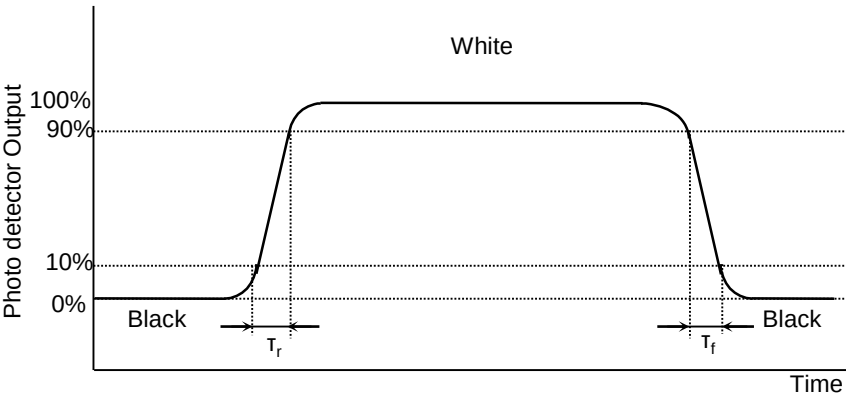


Fig 4.3 Viewing Angle Definitions

Product Specification

[Note 6]

Response time is obtained by measuring the transition time of photo detector output, when input signals are applied so as to make the area “black” to and from “white”.



[Note 7]

Fig 4.4 Response Time Definition

The brightness measurement is taken at point 13P/160P.

Brightness Uniformity = $\frac{\text{Minimum Photo detector output for P1-P13 with all pixels white}}{\text{Maximum Photo detector output for P1-P13 with all pixels white}} \times 100$

Brightness Uniformity = $\frac{\text{Minimum Photo detector output for P1-P160 with all pixels white}}{\text{Maximum Photo detector output for P1-P160 with all pixels white}} \times 100$

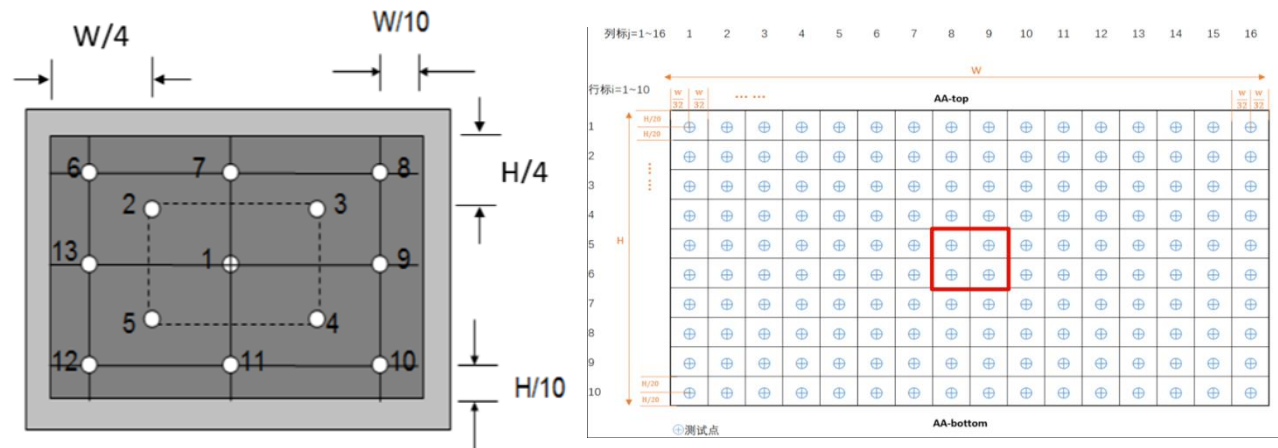


Fig 4.5 Brightness Measurement Points

Product Specification

5. MECHANICAL CHRACTERISTICS

The contents provide general mechanical characteristics for the model.
In addition the figures in the next page are detailed mechanical drawing of the LCD.

Items	Dimension	Typ.	Tolerance	Unit
Mother Glass	Size	1500*1850	-	mm
Q-Panel	Size	740*914	-	mm
C/F and TFT thickness after slimming		0.175		mm
Panel	A/A	118.464x189.5424		mm
	C/F	121.464x192.5424		mm
	TFT	121.464x195.2424		mm
	BM(U/D/L/R)	1.5/1.5/1.5/1.5		mm
	IC Bonding Area	2.7		mm
	Pol Size	CF:191.24x120.16 TFT:191.79x121.36	0.12	mm
	Gap Between Pol~Glass(U/D/L/R)	CF:0.65/0.65/0.65/0.65 TFT:0.05/3.4/0.05/0.05		mm
Module	Horizontal	125.96	0.05	mm
	Vertical	205.04	0.05	mm
	Thickness	2.15	0.20	mm
	UV Glue Thickness	0.80MAX	-	mm

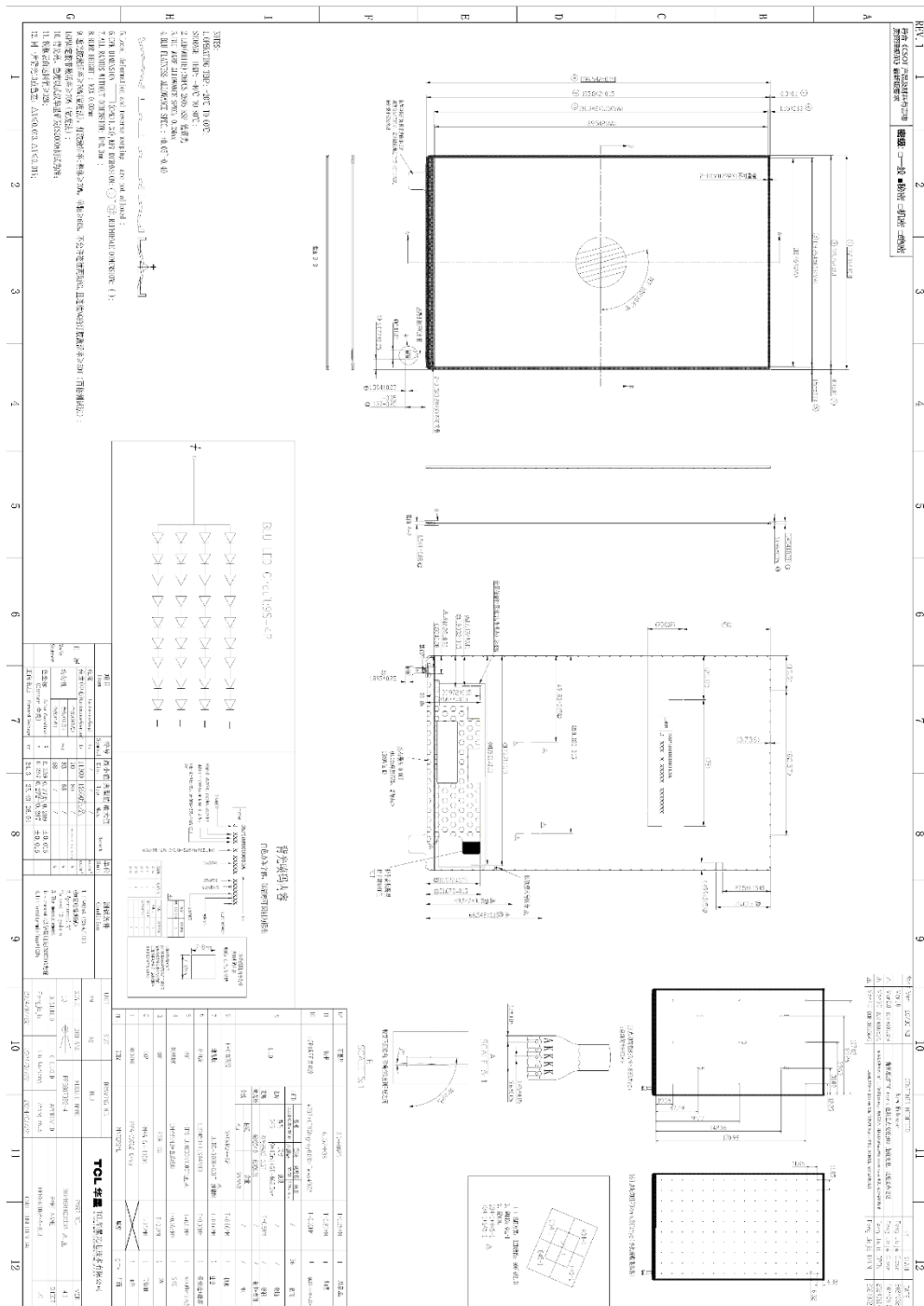
5.1 TLCD Drawing

Folded and unfolded status



Product Specification

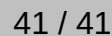
5.2 BLU Outline Dimension
Including each film



5.3 Panel Outline Dimension



5.4 FPCA Outline Dimension



Product Specification

6. RELIABILITY TEST

Must be accordance with Lenovo RA test items

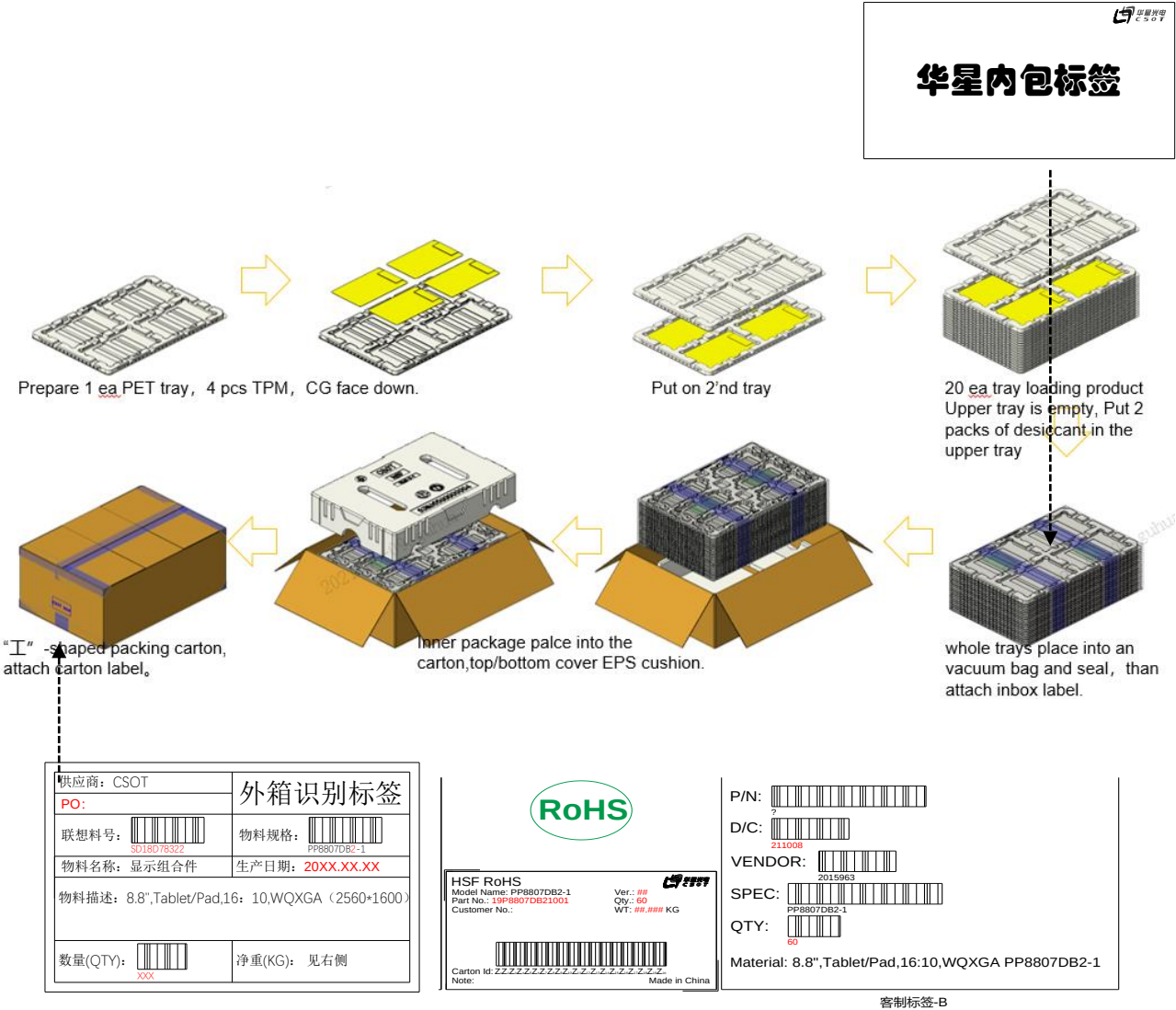
测试项目	测试条件
Image sticking	测试环境100±50lux，常温5pcs： 室温20±5°C，在棋盘格画面点亮30 min后，用128灰阶画面检查，常温残影现象需在30s内消失.
ESD	系统模型：150pf/330Ω；HBM模型：100pf/1500Ω： 1、不上电Contact测试，打引线，HBM模型Limit ±2.5KV，系统模型Limit ±2KV，每个点位放电次数5次。 2、上电测试，CG朝下，放电打铁框9点，每点打5次放电1次，系统模型；标准：Air Limit ±8KV，Contact Limit ±4KV 3、上电测试，CG朝上，放电打CG9点，每点10次放电1次，系统模型；标准：Air Limit ±15KV，Contact Limit ±8KV 结果以整机为准
高温驱动（HTO）	温度60°C，240h
低温驱动（LTO）	温度-20°C，240h
高温高湿驱动（HTBO）	温度60°C，湿度90%，240h
高温存储（HTS）	温度80°C，240h
低温存储（LTS）	温度-40°C，240h
高温高湿存储（HTBS）	温度60°C，湿度90%,240h
寿命测试（ALT）-工作	温度-10~60°C，湿度0~90%，240h（测试10cycle)
Thermal shock test	‘-40°C/1h->60°C/1h，共30cycle
小球跌落	直径Φ32mm，weight 130g,使用铝合金治具进行测试；在白画面下，将不锈钢球，AA区平面区域（除元器件区域）均分9官格，从16cm高度自由跌落，每片样品冲击9个点，每个点冲击1次，确认显示无异常（不允许CG/Panel破）
黄斑测试	直径Φ10mm的不锈钢球面压头，5mm/min，LCM向下放置在测试治具上，下方放置一面镜子（用于观察显示效果），测试点如图12点，要求显示屏在4N时显示正常，白画面显示无黄斑；
包装振动	依据《QT/WHCSOT-QU-PR-RA-014 振动测试技术标准》，1~200Hz,1.3Grms,30min/XYZ；
包装跌落	依据《QT/WHCSOT-QU-PR-RA-013 跌落测试技术标准》，一角三棱六面，81.3cm；

Product Specification

7. Package

7.1. Packing Description

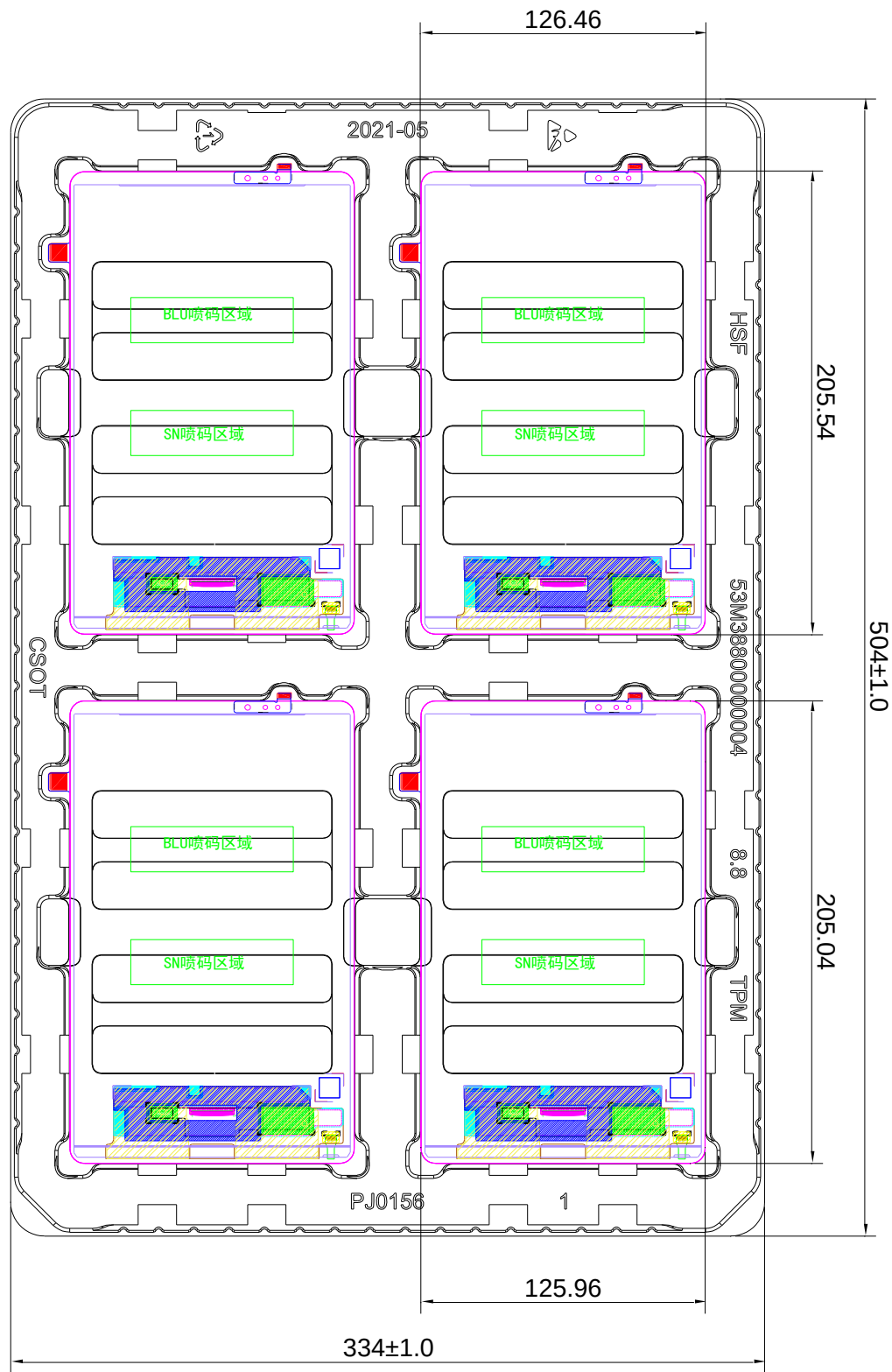
NO.	Description
1	PP8807DB2-1 TPM PET Tray, 504*334*13mm, 4
2	VMPET Vacuum Bag,794x384x0.09mm
3	EPS Cover,540x370x97mm
4	Carton,560x390x227mm
5	Label, Art Paper,80x40mm
6	Lenovo 8S label,9*9mm,PET,White
7	Lenovo label, 118*72mm, Art paper, white
8	Packing tape, W=60mm, 100M/ROA



PS : Red font content is variable

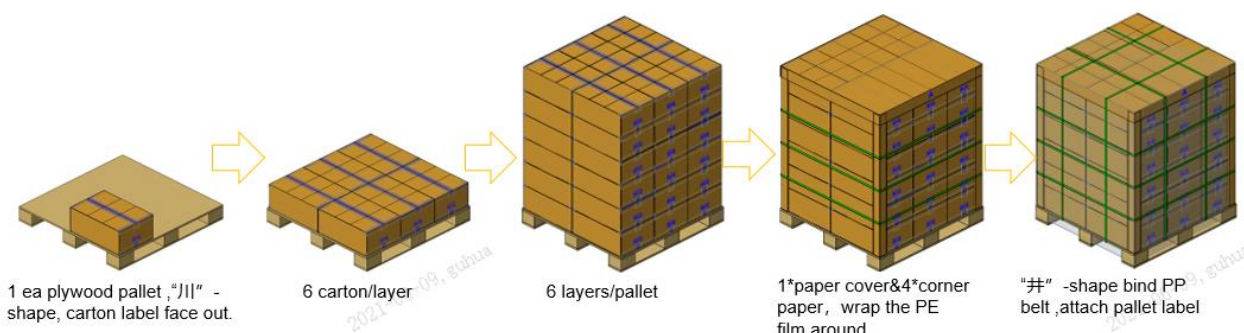
Product Specification

7.2. Description of Packing Tray



7.3. Description of Packing Box

NO.	Description
1	Plywood Pallet, 1190*1140*130mm, 川型
2	Paper Cover,1135x1185x100mm
3	Corner Paper3, 1320*50*50*5mm, Paper, For H
4	Label, Art Paper,80x40mm
5	PE Stretch Film,320mx500mmx0.02mm,3kg/Roll
6	PET Belt,Green,600m*16mm*0.8mm



供应商: CSOT	外箱识别标签
PO:	
联想料号:  SD18D78322	物料规格:  PP8807DB2-1
物料名称: 显示组合件	生产日期: 20XX.XX.XX
物料描述: 8.8",Tablet/Pad,16: 10,WQXGA (2560*1600)	
数量(QTY):  XXX	净重(KG): 见右侧

HSF RoHS

Model Name: PP8807DB2-1

Ver.: ##

Carton Qty.: 36

Qty.: 2160





Pallet Id:

Note:

Made in China

Product Specification

8. Incoming Inspection Standard

Reference<IIS>

Product Specification
9. Checklist

	Description	Request	Value
Connection	MIPI Impedance	100±10Ω	45±5Ω
	B2B CNT	FR4, 0.3mm	ZIF SUS
	Pin definition	LED+/- isn't next to TE/VCC/IOVCC	OK
	ID Pin	1 st source: GND 2 nd source: 1.8V(different cell) 2 nd source: GND(same cell+IC)	GND (需客户确认)
IC	Protection	Shielding tape on IC	NT36523NEH-DP/3YA
FPC	Bending Area	Not exceed M/F	Y
	FPC layer	Single layer FPC where add adhesive tape	NA
	FPC status	Unfolded while direct bonding	Y
	GND Area	Need GND area to connect SUS	Y
	Test Points	Need shieling tape on it	Y
	Bending Area	Single layer	Y
	Via Hole	2 holes on ending area, Φ>1.7mm	NA
SUS	4 Corner side	At lease 1.2mm	Y
Protective Film	Size	Length: 10mm+10mm; Width: 8mm	NA
Panel	Glass Generation	G5?G6?	G6
	Cutting Q'ty	Q-panel and panel Q'ty	96
	Display mode	VA/IPS	FFS
	Mask Q'ty	Array mask	10
	Pixel Domain	1 or 2?	2
	ITO@C/F	ITO square resistance, thickness	Conductive Pol
	Scan direction	Single scan or dual scan	Duan scan
	PS Parameter	Main/Sub PS density and size	MPS:8.6,0.088%; SPS:8.8,3.14%。
	Cell gap	Center point	2.8um
	LC injection	Vacuum injection or ODF	ODF
	LC Margin	>6%	Y
	Pol compensation	A+C/B+B/None compensation film	None compensation film
	UV Glue	Fill out at IC around	Y
	Pol surf. Treatment	Direct bonding: HC+Glare Air bonding: Haze44+glare(>4inch)	HC
	Pol position	Direct bonding: pol is higher 0.05mm than M/F Air bonding: pol is lower 0.05mm than M/F	Y
Package	Surface resistance	10 ⁴ ~10 ⁹ Ω	Y
	Friction voltage	≤100V	Y
	Layer in one Box	<10layer	Y